KARTHIKEYAN P

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LinkedIn | Github | Coursera

PROFESSIONAL SUMMARY

Generative AI Developer with nearly 3 years of experience building and deploying enterprise-scale LLM applications. Proven track record of leveraging advanced RAG pipelines, model fine-tuning, and cloud platforms (GCP, Azure) to deliver measurable business value, including a 15% reduction in unplanned downtime and a 20% increase in team productivity. Seeking to apply deep expertise in LLM architecture and multi-agent systems to solve complex business challenges.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL

AI/ML Frameworks & Libraries: LangChain, LangGraph, TensorFlow, PyTorch, Scikit-learn, CrewAI

- **Tools & Platforms:** Docker, Streamlit, FastAPI, Vertex AI, Google Cloud Platform (GCP), Azure ML, Azure DevOps, SAP HANA, SAP Analytics Cloud, Git, GitHub
- **Vector DBs:** Pinecone, ChromaDB, Neo4j (Graph)
- **Concepts:** Retrieval-Augmented Generation (RAG), LLMOps, Model Fine-Tuning (LoRA), Agentic AI & Multi-Agent Systems, Function Calling, Prompt Engineering, Data Wrangling

PROFESSIONAL EXPERIENCE

Maventic Innovative Solutions | Bengaluru, Karnataka
GenAI Developer | May 2022 – Present

- Engineered and deployed enterprise chatbots using LangGraph and FastAPI to automate over 500 daily queries to SAP HANA and CRM systems, reducing manual data retrieval time for the sales team by 10+ hours per week.
- Designed and implemented predictive maintenance models that achieved 95% accuracy in forecasting equipment failures, leading to a 15% reduction in unplanned downtime for key industrial assets.
- Led the architectural design of a core RAG pipeline, selecting Pinecone for vector storage and integrating it with a Neo4j knowledge graph to improve retrieval relevance for complex technical documents by 40%.
- Developed and deployed live, interactive dashboards using Streamlit, providing real-time model predictions and OEE metrics to over 25 engineering and management stakeholders.
- Managed the end-to-end model lifecycle using Azure ML Studio and Azure DevOps, establishing a CI/CD pipeline that automated model testing and deployment, reducing release cycles by 50%.

• Collaborated with product managers to translate business requirements into a technical roadmap and mentored two junior developers on best practices for prompt engineering and MLOps.

PERSONAL & ACADEMIC PROJECTS

AI Research Analyst Agent
Architected an autonomous research agent using LangGraph to perform multi-step web research and generate structured reports. Leveraged graph state for short-term task memory and a vector database for long-term knowledge caching.
Tech Stack: LangGraph, LangChain, Tavily API, Pinecone, Streamlit.

Personalized Coding Co-Pilot
Developed a context-aware coding assistant using a RAG pipeline by embedding an entire codebase into a ChromaDB vector store for long-term memory. Utilized ConversationBufferMemory for coherent, multi-turn dialogue.
Tech Stack: LangChain (LCEL), ChromaDB, Tree-sitter, OpenAI API, Streamlit.

Dynamic Customer Support Agent
Designed a stateful support agent with LangGraph capable of using custom LangChain Tools to query live databases (e.g., check order status). Integrated a dual-memory system for short-term conversational context and long-term user history.
Tech Stack: LangGraph, LangChain (Tools), FastAPI, SQL, Redis.

CERTIFICATIONS

- Machine Learning | Stanford University
- Data Science | IBM
- Prompt Designing | Google
- Streamlit with Gemini in VertexAI | Google Cloud Platform (GCP)
- RAG with VectorDB and KnowledgeGraph | Deeplearning.ai
- Function Calling and Data Extraction using LLM | Deeplearning.ai
- LangGraph | Langgraph
- LLMOps | Deeplearning.ai

EDUCATION

Woolf University (Alma Better) | Remote
Master of Science in Artificial Intelligence and Machine Learning | 2024 – Present
CGPA: 8.7

KCG College of Technology | Chennai, Tamil Nadu
Bachelor of Engineering in Computer Science and Engineering | 2019 – 2023
CGPA: 8.5